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A procedure for the evaluation of measurement uncertainty based on moments

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Abstract

A practical, transparent and accurate procedure is proposed for the evaluation of measurement uncertainty. The procedure is based on established statistical theory, allows the treatment of correlated input quantities estimated from a finite number of observations and permits the assignment of asymmetric distributions to input quantities.

The procedure is applicable wherever the basic method of clause 8 of the *Guide to the Expression of Uncertainty in Measurement* (the *Guide*) is applicable. As in the method of the *Guide*, distributions are associated with input quantities for use with a linearized formulation of the measurement function. These distributions are combined using a property of their statistical cumulants, which are simple functions of their moments. The distribution ultimately obtained for the measurand is a member of the Pearson family, which has the set of scaled *t*-distributions as a subset. So the procedure allows greater flexibility than the method of the *Guide* in the representation of the distribution associated with the measurand.

The procedure has the qualities of internal consistency and transferability as outlined in clause 0.4 of the *Guide* and is consistent with principles of the ‘propagation of distributions’ that underlie a proposed supplement to the *Guide*. In many cases it will permit the quotation of uncertainty intervals that are narrower than those of the *Guide* yet with the same justifiable probability of containing the measurand, so providing a more informative statement to the user of the interval. The procedure is presented for the probabilities of 0.95 and 0.99 but can be extended to accommodate other probabilities.

1. Introduction

Various methods have been proposed for evaluating the uncertainty to be attached to a measurement result [1–5], including some presented since the publication of the *Guide to the Expression of Uncertainty in Measurement* [6] (the *Guide*). Such methods reflect understandings of probability that accord somewhat with the conflicting frequentist and Bayesian schools of statistical inference, and possibly also with a theory not widely favoured known as fiducial inference [7–10]¹. The difficulties in implementing these understandings in practical

situations relate chiefly to the need to combine ‘random’ and ‘systematic’ errors appropriately.

A frequentist procedure for estimating the value of a measurand would likely involve constructing a confidence interval that would be applicable if all the systematic errors were zero, and then adding the extreme possible contributions from these errors to the limits of this interval. This appeal to worst-case values can seem to many to be an overly conservative way of incorporating systematic errors.

In contrast, a Bayesian procedure requires assigning a prior probability density function (pdf) to all unknown quantities, including the mean and variance of any distribution of repeated measurements. These pdfs represent the state of

¹ An interesting and potentially valuable approach called ‘generalized inference’ has recently been proposed by Wang and Iyer [11].

belief of the experimenter about these quantities before the experiment. The likelihood function of subsequent data is then used to update these prior pdfs to posterior pdfs, via analysis that can be summarized by the relationship

$$\text{posterior pdf} \propto \text{likelihood function} \times \text{prior pdf} \quad (1)$$

and then the posterior pdf of the measurand is calculated. This assignment of prior distributions to all influence quantities avoids the criticism of the frequentist method above. However, not everyone will necessarily agree with the choice of prior distributions, and exact calculation of the posterior distribution of the measurand involves an amount of computation that is impractical for routine use.

The practical procedure recommended by the *Guide* makes a distinction between Type A uncertainty components, which are those that can be evaluated and reduced by repeated measurement, and Type B uncertainty components, which are those that can be evaluated and reduced only by other means². In its basic procedure (which is summarized in its clause 8), Type A components are evaluated using the usual frequentist method but Type B components are assigned pdfs, as in a Bayesian analysis. The components are then combined with one another in a manner that derives its validity from frequentist theory [12, 13]. The overall procedure mixes frequentist and Bayesian concepts and is likely to be of concern to proponents of statistical propriety [14, 15].

In this paper, we describe a procedure for uncertainty assessment that avoids many of the above criticisms; the systematic errors are not represented by worst-case values, the procedure requires very little computation and the development follows established statistical theory. The procedure is applicable wherever the basic method of the *Guide* is applicable and satisfies the *Guide*'s criteria of internal consistency and transferability. As advocated in a proposed supplement to the *Guide* [16], distributions are assigned to input quantities. If these distributions are adequate representations of belief then the procedure gives an expanded uncertainty interval that contains the value of the measurand with a probability close to the quoted value, be it $p = 0.95$ or $p = 0.99$ or any other reasonable probability.

Section 2 presents the basic technique. Section 3 applies this in an example involving one Type A evaluation and one Type B evaluation and compares our results with those of other methods. Section 4 generalizes the method first by considering special types of Type A evaluation, second by considering the imprecise assessment of scale in Type B evaluation and third by allowing for correlation between input quantities, including those that are based on a finite number of repeated measurements. Section 5 shows how the method may be extended to allow asymmetric input distributions, which occur naturally in certain measurement situations. Section 6 addresses the issues of internal consistency and transferability that are important for any practical procedure. Section 7 comments comparatively on the method of the *Guide* and the proposed method. Various appendices contain much supporting material. Appendix A applies the method in

example H.1 of the *Guide* with notable results. Appendix B discusses performance criteria for a procedure generating an uncertainty interval. Appendix C comments on the Bayesian approach in this context. Appendices D–G demonstrate the validity of the aspects of the procedure.

2. Basic method

Consider a measurement function of m input quantities, $Y = F(X_1, \dots, X_m)$, and a measurement result, y , that is the same function of the estimates of the input quantities, $x_1, \dots, x_m$. Following the *Guide*, we approximate F in the region of the measurement result by a linear function

$$Y \approx y + E_Y, \quad E_Y = \sum_{i=1}^m c_i E_i, \quad E_i = X_i - x_i, \quad (2)$$

where c_i is equal to the value of $\partial F / \partial X_i$ at the point $(x_1, \dots, x_m)$. The E_i variables are deviations or 'error' variables that, for the moment, we regard as being independent and having mean zero³.

The pdf of Y is a translation of the pdf of the total error variable E_Y which itself is the m -fold convolution of the pdfs of the scaled E_i variables. In most situations it is difficult to derive a convolution of pdfs analytically, so a numerical method or simulation is required. As indicated, this computational requirement is a drawback of a standard Bayesian method—which has been followed up to this point. Such computation can be avoided by admitting an approximation to the distribution of Y , as permitted by the *Guide* in clause G.6.2. By finding an approximation to the distribution of E_Y we can calculate an interval containing approximately the central 100 p % of the distribution of Y , where $p = 0.95$ or $p = 0.99$ is the usual choice.

Let us make our convention of notation explicit. Quantities possessing pdfs, such as X_i and E_Y , are denoted in upper case to indicate that, in the context of the statistical analysis, they are regarded as random variables. Quantities that are fully known, such as the numerical values realized by random variables, are indicated in lower case, e.g. x_i and c_i . This notation is common in statistics. It enables the reader to distinguish between uncertain quantities, which are subject to probability statements such as $\Pr(X_i > 0) = 0.5$, and known quantities, which are not subject to (non-trivial) probability statements. In adherence to this convention, this paper uses T instead of the historical notation t to denote a variable with a Student t -distribution.

The method of approximating the distribution of E_Y involves the proper combination of the *cumulants* of the E_i variables—the set of first r cumulants being a transformation of the set of first r moments⁴. The first cumulant of any variable is its mean, the second cumulant is its variance, the third cumulant is its third central moment (which is zero for symmetric variables possessing three moments) and the fourth

³ More strictly, $-E_i$ is the error variable.

⁴ The cumulants for $r \geq 2$ are related to the central moments $\{\mu_r\}$ by $\kappa_r = \mu_r - \sum_{j=2}^{r-2} \binom{r-1}{j} \mu_j \kappa_{r-j}$ [17]. Like the r th moment about the origin and the r th central moment, the r th cumulant has the dimension of the measurement unit raised to the r th power.

² Strictly, it is the mode of evaluation that is called Type A and Type B; but see clauses 4.2.3 and 4.3.1 of the *Guide*.

Table 1. Variance u^2 and fourth cumulant w of symmetric input distributions.

Distribution	Variable	u^2	w
Laplace	$L(\delta)$	$2\delta^2$	$12\delta^4$
Scaled t	δT_ν	$\frac{\delta^2 \nu}{\nu - 2}$	$\frac{6\delta^4 \nu^2}{(\nu - 2)^2(\nu - 4)}$
Normal	$N(0, \delta^2)$	δ^2	0
Triangular	$I[-\delta, \delta]$	$\frac{\delta^2}{6}$	$-\frac{\delta^4}{60}$
Uniform	$U[-\delta, \delta]$	$\frac{\delta^2}{3}$	$-\frac{4\delta^4}{30}$
Arc-sine	$A[-\delta, \delta]$	$\frac{\delta^2}{2}$	$-\frac{3\delta^4}{8}$

cumulant is its fourth central moment minus three times the square of its variance.

The method is based on the principle that cumulants are summed when independent variables are summed [18, p 45]. That is, if $\kappa_r(\cdot)$ denotes the r th cumulant of the argument variable then

$$\kappa_r\left(\sum_j a_j Z_j\right) = \sum_j a_j^r \kappa_r(Z_j) \quad (3)$$

for any sets of independent random variables $\{Z_j\}$ and constants $\{a_j\}$. In particular, we will apply (3) with the variance and fourth cumulant, which we denote by u^2 and w , respectively. Until section 5, we assume that all input distributions are symmetric, so their third cumulants are zero.

All cumulants except the first are unaffected by a shift in mean. Therefore, the variance and fourth cumulant of X_i , which are u_i^2 and w_i , are also the variance and fourth cumulant of E_i . So, in that context, we can refer to the distributions of X_i and E_i interchangeably. The variance and fourth cumulant for various symmetric zero-mean input distributions are listed in table 1. These distributions are the Laplace distribution with standard deviation δ and variable denoted $L(\delta)$ ⁵; the t -distribution with ν degrees of freedom, with variable T_ν , scaled by a factor δ ; the normal distribution with zero mean and variance δ^2 , with variable $N(0, \delta^2)$; the isosceles triangular distribution on the interval from $-\delta$ to δ , with variable denoted $I[-\delta, \delta]$; the uniform distribution on the same interval, with variable $U[-\delta, \delta]$; and the arc-sine distribution on the same interval, with variable denoted $A[-\delta, \delta]$ ⁶. In each case δ is simply a scale factor.

Initially, we consider the variables $E_1, \dots, E_m$ to be independent, so from (3) the variance and fourth cumulant of E_Y are

$$u_Y^2 = \sum_{i=1}^m c_i^2 u_i^2, \quad w_Y = \sum_{i=1}^m c_i^4 w_i. \quad (4)$$

From these we calculate the dimensionless ratio

$$\gamma_Y = \frac{w_Y}{u_Y^4} \quad (5)$$

⁵ The variable $L(\delta)$ has pdf $f(x) = \exp(-|x|/\delta)/(2\delta)$ for $-\infty < x < \infty$.

⁶ The variable $A[-\delta, \delta]$ has the distribution of $\delta \sin \Phi$ where Φ is a uniformly distributed random phase. The corresponding pdf is $f(x) = \pi^{-1}(\delta^2 - x^2)^{-1/2}$ for $|x| < \delta$ and the cumulative distribution function is $\Pr(A[-\delta, \delta] \leq x) = \pi^{-1}\{\arcsin(x/\delta) + \pi/2\}$ for $|x| < \delta$ [19].

called the coefficient of *excess*, which is a function of the *shape* of the distribution of E_Y . The coefficient of excess for a general variable, $\gamma \equiv w/u^4$, refers to the excess of kurtosis (heaviness of the tail) over that of a normal distribution. It is zero for a normal distribution, greater than zero for a symmetric distribution with heavier tails, such as a t -distribution, and less than zero for a symmetric distribution with blunter tails, such as a uniform distribution⁷.

The distribution of E_Y can be approximated by another symmetric distribution with the same mean (zero), the same variance u_Y and the same coefficient of excess γ_Y ⁸. We define our approximation to be the corresponding unique distribution from the Pearson system of distributions⁹. If $\gamma_Y > 0$ this is a scaled t -distribution, if $\gamma_Y = 0$ it is a normal distribution, and if $\gamma_Y < 0$ it is a shifted and scaled symmetric beta distribution¹⁰. An indication of the shapes possible for the approximation can be obtained from figure 1, which shows various standardized symmetric Pearson distributions, i.e. those with mean 0 and variance 1. In particular, $\gamma = 6$ gives a scaled t -distribution with 5 degrees of freedom, $\gamma = -1.2$ gives a uniform distribution, and $\gamma = -1.5$ gives an arc-sine distribution.

The required half-width, h_p , of the 100

% uncertainty interval is the appropriate percentile of the approximating distribution. This is given by

$$h_p = k_p(\gamma_Y)u_Y, \quad (6)$$

$$k_{0.95}(\gamma) \approx \frac{1.96 + 1.845\gamma + 0.47\gamma^2}{1 + 0.906\gamma + 0.239\gamma^2}, \quad -1.2 \leq \gamma \leq 6,$$

$$k_{0.99}(\gamma) \approx \frac{2.5758 + 2.6736\gamma + 0.7685\gamma^2}{1 + 0.8864\gamma + 0.2362\gamma^2}, \quad -1.2 \leq \gamma \leq 6, \quad (7)$$

where (6) and (7) are close approximations to the relevant percentiles of the standardized symmetric Pearson distribution. These approximations have relative error within $\pm 0.025\%$ for $\gamma \in \{-1.2, 0.01, 6\}$, i.e. for symmetric Pearson distributions with shapes between that of the uniform distribution and the t -distribution with 5 degrees of freedom¹¹. (The leading term in each approximation is seen to be the appropriate percentile of the standard normal distribution.) The expanded uncertainty interval for the measurand Y is then given by $[y - h_p, y + h_p]$.

⁷ The coefficient of excess is often written as γ_2 , with γ_j for $j \neq 2$ being used to denote related coefficients. We have dropped the subscript 2 to simplify notation. The coefficients of excess for the distributions in table 1 in order are decreasing, namely $3, 6/(\nu - 4), 0, -0.6, -1.2$ and -1.5 . Note that the Laplace distribution has the same coefficient of excess as the t -distribution with 6 degrees of freedom.

⁸ In effect, a distribution is being obtained with the same first four moments, the first and third moments being zero. The approximation of distributions by matching moments is a standard statistical technique.

⁹ The Pearson system is a four-parameter system containing distributions with pdf $f(x)$ satisfying $df(x)/dx = (x - a_1)f(x)/(a_2 + a_3x + a_4x^2)$ for constants a_1, a_2, a_3 and a_4 . The system includes all linear transformations of most of the well-known continuous distributions, including normal, t , chi-square, uniform, gamma, beta and F -distributions [20]. Of the input distributions in table 1, the Laplace and triangular distributions are the only distributions that are not members of the Pearson system.

¹⁰ A symmetric beta distribution with shape parameter q has pdf $f(x) = B(q, q)^{-1}x^{q-1}(1-x)^{q-1}$ on the interval $[0, 1]$ and has mean 0.5. Here $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ is the beta function.

¹¹ Equations (6) and (7) were obtained for this paper by an optimization method involving rational functions, i.e. the ratio of polynomials. The emphasis in this process was on minimizing the maximum error over a realistic range of values of γ while maintaining the approximations in a simple form.

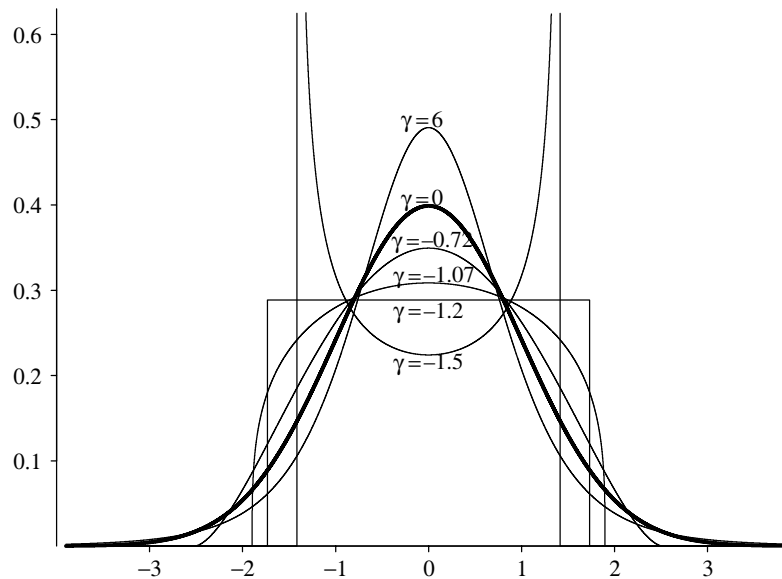


Figure 1. Probability density functions of symmetric Pearson distributions with mean 0, variance 1 and coefficient of excess γ . The standard normal pdf ($\gamma = 0$) is highlighted.

3. Example using Type A and Type B evaluation

Consider, for example, the simple measurement function $Y = X_1 + X_2$, with X_1 being estimated by averaging repeated measurements and X_2 being estimated by other means. In accordance with the principles of Bayesian statistics, X_1 and X_2 are each assigned a pdf before making any repeated observations.

Suppose that estimating X_1 involves the averaging of n observations assumed to be drawn independently from a normal distribution with mean X_1 and unknown variance Σ^2 . Our knowledge of X_1 and Σ^2 is initially expressed in terms of a prior joint pdf. The joint pdf that is usually chosen when nothing is known about X_1 and Σ^2 before acquiring the sample is

$$f_{X_1, \Sigma^2}(x, z) \propto \frac{1}{z}, \quad (8)$$

which is the product of the independent prior (improper) densities for the mean and variance given by $f_X(x) = \text{constant}$ and $f_{\Sigma^2}(z) \propto 1/z$. (Here x and z are simply mathematical dummy variables.) A clear justification of this choice of joint pdf in the absence of other knowledge is given by Jaynes [21]. However, we make some cautionary comments about using (8) in section 8¹².

The sample of observations $\{x_{[1]}, \dots, x_{[n]}\}$ provides information with which we can update our knowledge of X_1 and Σ^2 using (1). The sample mean is $\bar{x} \equiv \sum x_{[i]}/n$ and the sample variance is $s^2 \equiv \sum (x_{[i]} - \bar{x})^2/(n-1)$. If (8) describes the prior belief then the updated (i.e. posterior) pdf of X_1 is that of a variable with the t -distribution with $n-1$ degrees of

freedom, T_{n-1} , scaled by $s/\sqrt{n}$ and shifted by $\bar{x}$. That is, after the data are observed

$$X_1 \sim \bar{x} + \frac{T_{n-1}s}{\sqrt{n}}, \quad (9)$$

where $\sim$ can be read as ‘is distributed as’.

Equation (9) is a standard result in Bayesian statistics, see, e.g. [22, p 70; 23, p 69]. Kacker and Jones [15] have suggested that this Type A evaluation be treated in this way to help the procedure of the *Guide* to be interpreted from a Bayesian viewpoint. A proposed supplement to the *Guide* [16, section 4.1] also advocates the choice of (9) to represent the distribution of X_1 in this situation.

Suppose that the value of the second input quantity is known to lie in some interval $[b-0.5, b+0.5]$, with no value being regarded as more probable than any other. Then we treat X_2 as a variable with a uniform distribution on that interval.

The estimates x_1 and x_2 are taken to be the means of the distributions of X_1 and X_2 , so $x_1 = \bar{x}$ and $x_2 = b$ and the measurand $Y = X_1 + X_2$ can be written as

$$Y = \bar{x} + b + E_Y, \quad E_Y = E_1 + E_2 \sim \frac{T_{n-1}s}{\sqrt{n}} + U[-0.5, 0.5].$$

We find the variances and fourth cumulants of E_1 and E_2 from table 1 (or from first principles), and calculate $u_Y^2 = u_1^2 + u_2^2$ and $w_Y = w_1 + w_2$. Subsequently we find γ_Y from (5) and the half-width h_p from (6) or (7) (presuming that $-1.2 \leq \gamma_Y \leq 6$).

Let us call the method the ‘cumulants’ method. The half-widths of the intervals calculated by this method for various n and s are shown in the second column in the body of table 2. The other columns give the half-widths for intervals derived using a Bayesian procedure, a frequentist procedure and the procedure of the *Guide*. These procedures are described later. In every case the interval is given by $[\bar{x} + b - h, \bar{x} + b + h]$, where h is the quantity tabulated.

¹² This distribution is known as the Jeffreys prior for this situation, following the work of Jeffreys on choosing prior distributions with certain invariance properties. Note that the corresponding prior pdf of each of the alternative dispersion parameters $1/\Sigma^2$, Σ and $1/\Sigma$ has the same ‘ $1/z$ ’ form, so the joint densities $f_{X_1, 1/\Sigma^2}(x, z) \propto 1/z$, $f_{X_1, \Sigma}(x, z) \propto 1/z$ and $f_{X_1, 1/\Sigma}(x, z) \propto 1/z$ each describe the same state of belief as (8).

Table 2. Half-widths of 95% and 99% uncertainty intervals. The figure $r = (s/\sqrt{n})\sqrt{(n-1)/(n-3)} \div \sqrt{1/12}$ is the ratio of the standard deviation of X_1 to that of X_2 .

n	r	95%				99%			
		Bayesian	Cumulants	Frequentist	Guide	Bayesian	Cumulants	Frequentist	Guide
6	0.1	0.48	0.48	0.56	0.57	0.51	0.50	0.59	0.75
6	0.3	0.51	0.53	0.67	0.58	0.60	0.59	0.77	0.76
6	1	0.77	0.81	1.07	0.74	1.08	1.17	1.40	0.99
6	3	1.80	1.82	2.22	1.73	2.77	2.83	3.20	2.55
6	10	5.77	5.78	6.25	5.74	9.04	9.05	9.52	8.94
12	0.1	0.48	0.48	0.56	0.57	0.52	0.50	0.58	0.75
12	0.3	0.52	0.53	0.67	0.59	0.60	0.58	0.74	0.77
12	1	0.78	0.80	1.07	0.78	1.03	1.04	1.31	1.04
12	3	1.81	1.81	2.22	1.79	2.52	2.53	2.93	2.48
12	10	5.77	5.78	6.25	5.77	8.14	8.14	8.61	8.12

3.1. Comparison with other intervals

A standard Bayesian method (using so-called ‘non-informative’ prior distributions) is the same as the cumulants method except that the distribution of E_Y is evaluated to arbitrary precision and not approximated by the Pearson distribution. The corresponding figures differ very little from the figures for the cumulants method except where the standard deviations of X_1 and X_2 are comparable.

The usual frequentist procedure in this problem would regard X_1 , Σ^2 and X_2 as unknown constants about which we knew nothing except that X_2 lay in the interval $[b - 0.5, b + 0.5]$. Then, as is well known, the unknown quantity $(\bar{x} - X_1)/(s/\sqrt{n})$ is the realization of a variable with the t -distribution with $n - 1$ degrees of freedom. So an exact 95% confidence interval for X_1 is

$$\left[\bar{x} - \frac{t_{95,n-1}s}{\sqrt{n}}, \bar{x} + \frac{t_{95,n-1}s}{\sqrt{n}} \right], \quad (10)$$

where $t_{95,v}$ is the $(100 + 95)/2 = 97.5$ percentile of the t -distribution with v degrees of freedom. The correct interpretation of this is that the sampling process has a probability of exactly 0.95 of generating an interval containing X_1 no matter what the values of the unknowns X_1 and Σ^2 . So, to ensure that the process has probability of at least 0.95 of generating an interval containing Y no matter what the value of X_2 also, the extreme possible values of X_2 are added to the appropriate limits of this confidence interval. The result is the (statistically) conservative 95% confidence interval for Y , $[\bar{x} + b - t_{95,n-1}s/\sqrt{n} - 0.5, \bar{x} + b + t_{95,n-1}s/\sqrt{n} + 0.5]$. The analogous conservative 99% confidence interval is obtained by replacing $t_{95,n-1}$ with $t_{99,n-1}$. The adjective ‘conservative’ in statistics means that the probability is at least the stated amount and implies that it is likely to exceed that amount. This is consistent with the results of table 2 where, in general, the widths of the frequentist intervals are larger than those calculated by the other methods¹³.

If applied in this simple situation, the procedure of (clause 8 of) the *Guide* would make no reference to the prior distributions of X_1 and Σ , and would describe the associated uncertainty component by the measure of spread $s/\sqrt{n}$ and the number of degrees of freedom $n - 1$ (in accordance with

a Type A evaluation). The uncertainty component associated with the distribution of X_2 would be described by the standard deviation $1/\sqrt{12}$ and an infinite number of degrees of freedom, which represents certainty about the identities of the limits $b - 0.5$ and $b + 0.5$ (in accordance with a Type B evaluation). The two components would be combined using the so-called Welch–Satterthwaite formula¹⁴ to obtain the squared standard uncertainty and effective degrees of freedom, $u_c^2 = s^2/n + 1/12$ and $\nu_{\text{eff}} = (n - 1)n^2 u_c^4 / s^4$. The 95% interval would be $[\bar{x} + b - u_c t_{95,\nu_{\text{eff}}}, \bar{x} + b + u_c t_{95,\nu_{\text{eff}}}]$, while the 99% interval would use $t_{99,\nu_{\text{eff}}}$. In table 2 some of the figures for this method differ considerably from those for the Bayesian and cumulants methods, in particular where r , the ratio of standard deviations, is small enough for the uniform variable to dominate.

This example illustrates the accuracy of the cumulants method (if the Bayesian method is regarded as being the theoretical method of choice). In practice there will be more variables and the central limit theorem will take effect. So we can expect even greater accuracy because the limiting, normal, distribution is a member of the Pearson system.

Both the proposed interval and the interval of the *Guide* are defined using three quantities that correspond to the location, scale and shape of a symmetric distribution. In both methods the measure of location is the measurement estimate, $\bar{x} + b$. In the cumulants method the scale is described by u_Y and in the method of the *Guide* by u_c . In the cumulants method the shape measure, γ_Y , can be positive or negative. So the distribution of the measurand can have the form of either a t -distribution or a symmetric beta distribution. However, in the method of the *Guide* the shape measure is the positive figure ν_{eff} , so the distribution from which the interval is calculated is necessarily a t -distribution. The inability of the method of the *Guide* to involve a symmetric beta distribution is the chief reason for its poor performance in table 2 at low values of r . The greater flexibility of the cumulants method is also evident in the more realistic example given in appendix A.

Our interpretation of the relative performances is based on the premise that the Bayesian interval is the ‘best’ interval, which is a premise that we must accept if the Bayesian framework is adopted with the Jeffreys prior representing the

¹³ Note that ‘wide’ intervals do not guarantee that the procedure is conservative. See appendix B.

¹⁴ The Welch–Satterthwaite formula is an approximate frequentist technique for estimating the sum of the means of several normal random variables with unknown variances. So the Type B component is essentially being treated as if it arose from a very large sample of normally distributed repeated measurements [12, 13].

prior belief. We have more to say about the quality of an uncertainty interval (more strictly, the procedure generating the interval) in [appendix B](#). There are additional comments made about the Bayesian approach in [appendix C](#).

4. General method

Now we describe further aspects of the method that, together, make the general procedure widely applicable. Section 4.1 discusses Type A evaluation when the variance has been estimated from a separate set of observations, and section 4.2 presents a modification for Type A evaluation with low degrees of freedom, where the fourth cumulant or variance can be undefined. Section 4.3 allows for imprecise identification of the scale of an input distribution in Type B evaluation. Section 4.4 extends the method to allow correlation between input quantities, i.e. correlation between the E_i variables, even for small samples. Section 4.5 gives a brief summary of the full procedure.

4.1. Type A evaluation with variance estimated separately

The method of Type A evaluation giving rise to the posterior distribution (9) requires the variance to be estimated from the same sample that was used in the measurement. However, sometimes the variance will have been estimated from a previous, separate sample of size k , say. In that case the posterior distribution of the relevant input quantity X_i is

$$X_i \sim \bar{x} + \frac{T_{k-1} s}{\sqrt{n}}. \quad (11)$$

This result is consistent with (9), as we might have hoped, but here $\bar{x}$ is calculated from the sample of size n and s^2 is calculated from the sample of size k . This situation is not treated explicitly in the standard texts observed, so we give a brief derivation of (11) in [appendix D](#). This result is used in the example in [appendix A](#).

4.2. Type A evaluation with four or fewer degrees of freedom

As described so far, the procedure seems inappropriate when any error variable E_i is a multiple of a t -variable with $\nu \leq 4$, because the fourth cumulant of such a variable is undefined. The fourth cumulant could be treated as infinite but w_Y would become infinite no matter how minor this component of uncertainty was, and the shape of the distribution assigned to E_Y would be that of a t -distribution with four (or fewer) degrees of freedom. The problem would be worse if the component had $\nu = 1$ or $\nu = 2$ because u_Y^2 would then be infinite, and we would lose all sense of scale.

One way to avoid this problem is to base all Type A evaluations on a sample of six or more observations, but this may be impractical and expensive. An alternative is to use a suitably different prior distribution for X_i in the relevant Type A evaluation. The prior pdf for X_i contributing to (8) is the improper uniform density $f_{X_i}(x) = \text{constant}$, which does not place any limits on the range thought possible for X_i . To some degree, this pdf must misrepresent actual prior belief because it is non-zero at unrealistic values for X_i , such as positive values that are obviously orders of magnitude too large

Table 3. Variance u^2 and fourth cumulant w assigned to variables with scaled t -distributions with four or fewer degrees of freedom.

Variable	u^2	w
δT_1	$27 \delta^2$	$109\,350 \delta^4$
δT_2	$6.75 \delta^2$	$1708.593\,75 \delta^4$
δT_3	$3 \delta^2$	$150 \delta^4$
δT_4	$2 \delta^2$	$37.5 \delta^4$

to be reasonable and negative values for a quantity that might obviously be positive. It is only because of this unrealistic infinite range that infinite moments can arise.

The sample size and choice of prior distribution are under the influence of the metrologist, so this potential problem of $\nu \leq 4$ can be addressed during the phase of acquiring the input uncertainty data¹⁵. However, there still seems to be a need for a practical modification of the procedure in case such variables remain. The approach we take is to substitute certain finite values in place of the infinite moments, while ensuring that u^2 , w and γ remain continuous functions of ν . (Note that t -distributions can also be defined for non-integral values of ν .)

So variables with $\nu < 5$ are assigned a value of $\gamma = 150/\nu^2$, which gives continuity at $\nu = 5$ where $\gamma = 6$. Also variables with $\nu < 3$ are assigned a value of $u^2 = 27/\nu^2$, which gives continuity at $\nu = 3$ where $u^2 = 3$. The resulting exact figures of u^2 and w are found in table 3.

These figures are used where appropriate in (4) in the calculation of u_Y^2 , w_Y and γ_Y . If we find that $\gamma_Y \leq 6$ then we proceed as earlier with (6) and (7), in which case E_Y is being treated as a scaled t -variable with $\nu_Y = 4 + 6/\gamma_Y$ degrees of freedom if γ_Y is positive. Alternatively, if $\gamma_Y > 6$ then we undo part of our approximation by treating E_Y as a scaled t -variable with $\nu_Y = \sqrt{150/\gamma_Y}$ degrees of freedom. Then if we find that $\nu_Y < 3$ we undo another part of the approximation by treating the variance of the corresponding t -distribution as if it were $27/\nu_Y^2$. So if $\gamma_Y > 6$ we calculate

$$\nu_Y = \sqrt{\frac{150}{\gamma_Y}},$$

$$k_p(\gamma_Y) = \begin{cases} t_{100p, \nu_Y} \times \sqrt{\frac{v_Y^2}{27}} & \nu_Y < 3, \\ t_{100p, \nu_Y} \times \sqrt{\frac{\nu_Y - 2}{\nu_Y}} & 3 \leq \nu_Y \leq 5, \end{cases}$$

$$h_p = k_p(\gamma_Y) u_Y$$

and the expanded uncertainty interval is $[y - h_p, y + h_p]$. The equations

$$t_{95, \nu} \approx \frac{1.96 + 0.3246/\nu + 1.1405/\nu^2}{1 - 1.0318/\nu + 0.3014/\nu^2}, \quad 1 \leq \nu \leq 5, \quad (12)$$

$$t_{99, \nu} \approx \frac{2.5758 + 0.78/\nu + 3.607/\nu^2 + 1.505/\nu^3}{1 - 1.594/\nu + 0.918/\nu^2 - 0.191/\nu^3}, \quad 1 \leq \nu \leq 5, \quad (13)$$

¹⁵ Yet if the prior distribution truly represents the state of prior belief then there is no choice in the matter.

give approximations to the 97.5 and 99.5 percentiles of the t -distribution with relative error within $\pm 0.03\%$ for $\nu \in \{1, 0.01, 5\}$. (These complement the approximations (6) and (7) that were given earlier for $-1.2 \leq \gamma_Y \leq 6$.)

Appendix E contains further discussion of this modification and an example illustrating its performance.

4.3. Imprecise limits in Type B evaluation

The *Guide* recognizes that the limits of the distribution associated with a Type B evaluation might not be known exactly. Its procedure associates a finite number of degrees of freedom with the relevant standard uncertainty for inclusion in the Welch–Satterthwaite formula. In the cumulants method the appropriate way to include such components is via the concept of a *compound distribution* (or *infinite mixture distribution*) [24, 25] as described now.

Suppose that the distribution of a general variable X conditional on the value of an uncertain parameter Θ is described by the pdf $f_{X|\Theta}(x)$. (The notation $|$ means ‘conditional on’.) Suppose also that the pdf of Θ is $f_{\Theta}(\theta)$. Then the unconditional pdf of X is the (compound) pdf $f_X(x) = \int_{-\infty}^{\infty} f_{X|\Theta}(x) f_{\Theta}(\theta) d\theta$, which is a weighted integral of the conditional pdfs¹⁶. Any unconditional moment of X about the origin is given by the same weighted integral of that conditional moment. So if the conditional pdf $f_{X|\Theta}(x)$ has mean zero for all Θ , and has variance $u_{X|\Theta}^2$ and fourth cumulant $w_{X|\Theta}$ then the unconditional variance of X is $u_X^2 = \int_{-\infty}^{\infty} u_{X|\Theta}^2 f_{\Theta}(\theta) d\theta$ and the unconditional fourth cumulant of X is $w_X = \int_{-\infty}^{\infty} \{w_{X|\Theta} + 3u_{X|\Theta}^4\} f_{\Theta}(\theta) d\theta - 3u_X^4$.

This can be applied, in particular, to a zero-mean variable E_i with a scale factor Θ that is known only to be within $\pm 100\epsilon\%$ of some value δ . In this case Θ is regarded as a uniform variable on the interval $[(1 - \epsilon)\delta, (1 + \epsilon)\delta]$, i.e.

$$\Theta \sim U[(1 - \epsilon)\delta, (1 + \epsilon)\delta]. \quad (14)$$

(So $f_{\Theta}(\theta) = 1/(2\epsilon\delta)$ for θ in this interval.) If the conditional variable is $L(\Theta)$, $N(0, \Theta^2)$, $I[-\Theta, \Theta]$, $U[-\Theta, \Theta]$ or $A[-\Theta, \Theta]$, and if u^2 and w denote the relevant entries in table 1, then the resulting variance and fourth cumulant of E_i are

$$u_i^2 = \left(1 + \frac{\epsilon^2}{3}\right) u^2, \quad (15)$$

$$w_i = \left(1 + 2\epsilon^2 + \frac{\epsilon^4}{5}\right) w + \left(4\epsilon^2 + \frac{4\epsilon^4}{15}\right) u^4. \quad (16)$$

These equations describe adjustments that can be applied to *any* zero-mean variable with a uniformly-distributed scale factor, including the t -variables of tables 1 and 3. However, the t -variable will tend to occur only in the manner described earlier, where the scale factor $s/\sqrt{n}$ is fully known.

¹⁶ When a distribution is formed as a weighted sum of a finite number of densities we obtain a *finite mixture* distribution, as in [26, 27]. In our case, there are an infinite number of contributing densities differing only through the value of θ .

4.4. Correlated input quantities

Modifications are required when there are correlations between the E_i variables, because (3) and, hence, (4) hold only if the variables are independent. One kind of correlation between the E_i variables occurs when two or more input quantities are assessed using the same estimate of a shared influence quantity. This has been well called a ‘logical correlation’. As implied in clauses 5.2.4, 5.2.5 and F.1.2.4 of the *Guide*, this kind of correlation can be dealt with by prescribing that such influence quantities be treated as input quantities themselves. For example, the measurement function might be $Y = F(X_1, X_2)$ with X_1 and X_2 both being assessed using the same estimate of X_3 , according to $X_1 = G(X_3, X_4)$ and $X_2 = H(X_3, X_5)$, say, where X_3 , X_4 and X_5 are independent. Then we formulate the measurement function as $Y = \tilde{F}(X_3, X_4, X_5)$ for the appropriate new function $\tilde{F}$. In this way, we can take such correlations into account without necessitating a change to (4).

The requirement that logical correlations be treated in this way may seem unreasonable because such decomposition of the measurement function may be difficult. However, it appears to be the only way in which these correlations can be taken fully into account. Often a single figure of correlation r_{ij} between two error variables E_i and E_j might be known. This gives us information about the expectation of the second-order term $E_i E_j$, which enables us to properly adjust the total variance u_Y^2 . However, it does not give us full information about the expectation of fourth-order terms such as $E_i^2 E_j^2$, which would be needed to properly adjust the shape parameter γ_Y (or the shape parameter ν_{eff} in the procedure of the *Guide*). Only if the relevant joint distribution were the bivariate normal distribution would r_{ij} be sufficient information for that purpose.

Another kind of correlation arises when more than one input quantity is assessed in the same set of repeated measurements. This can be called a ‘statistical correlation’. The method properly treats statistical correlations by obtaining exact figures for the variance and fourth cumulant of the contribution from each group of input quantities that were assessed in a single set of independent measurements. Subsequently, these figures contribute to the total variance and fourth cumulant through a generalization of (4). It is assumed that the vector of correlated observations in any measurement is drawn from a single multivariate normal distribution whose mean vector $\mathbf{X}$ and covariance matrix $\mathbf{V}$ have the prior joint density function

$$f_{\mathbf{X}, \mathbf{V}^{-1}}(\mathbf{x}, \mathbf{v}^{-1}) \propto |\mathbf{v}|, \quad (17)$$

which can be shown to be a generalization of (8). The resulting adjustment to the procedure is straightforward. This adjustment is stated here and derived in appendix F, where (17) is also explained further.

Recall that our input variables are indexed from 1 to m . Consider a subset of indices A for a set of variables estimated from the same set of n repeated observations. Let $x_{i,[k]}$ be the k th observation of X_i for $i \in A$. The quantities

$$\bar{x}_i = \frac{1}{n} \sum_{k=1}^n x_{i,[k]} \quad \text{and} \quad s_{ij} = \frac{1}{n-1} \sum_{k=1}^n (x_{i,[k]} - \bar{x}_i)(x_{j,[k]} - \bar{x}_j) \quad (18)$$

are the sample mean for X_i and sample covariance for X_i and X_j . We define

$$u_{ij} \equiv \frac{(n-1)s_{ij}}{(n-3)n}, \quad (19)$$

which gives $u_{ii} = u_i^2$. The contribution of the subset A to u_Y^2 and w_Y is

$$u_A^2 = \sum_{i,j \in A} c_i c_j u_{ij} \quad \text{and} \quad w_A = \frac{6u_A^4}{n-5}. \quad (20)$$

Also, suppose there are q such subsets of input variables, $A_1, \dots, A_q$ (which will generally have different sample sizes $n_1, \dots, n_q$). We define $A_0 \equiv A_1 \cup \dots \cup A_q$. Then (4) is to be replaced by

$$\begin{aligned} u_Y^2 &= \sum_{k=1}^q u_{A_k}^2 + \sum_{i \notin A_0} c_i^2 u_i^2, \\ w_Y &= \sum_{k=1}^q w_{A_k} + \sum_{i \notin A_0} c_i^4 w_i, \end{aligned} \quad (21)$$

where i runs from 1 to m . When $q = 0$ this reduces to (4).

The requirement that the underlying covariance matrix $\mathbf{V}$ be given a prior distribution means that the covariance between each pair of variables in any subset A_k is being given a prior distribution. Consequently, the result obtained if some sample covariance, s_{ij} , happens to be zero is not the result that would be appropriate if E_i and E_j were initially assumed to be independent. If we had been able to assume *a priori* that the true covariance of E_i and E_j was zero then X_i or X_j should have been removed from this subset of variables and treated independently.

4.5. Summary of general procedure

The general method for the assessment of uncertainty can be summarized as follows. The measurement function is expressed in terms of input quantities $X_1, \dots, X_m$ in which the only dependences arise from quantities being assessed in a common set of repeated measurements. The measurement function $Y = F(X_1, \dots, X_m)$ is then ‘linearized’ so that the difference between the measurand and the measurement estimate is the variable E_Y given in (2). The variances and fourth cumulants of the scaled contributions $c_1 E_1, \dots, c_m E_m$ are obtained from tables 1 and 3 and equations (15) and (16). These are combined using equations (18) to (21), the last of which reduces to (4) when the input quantities are measured independently. The coefficient of excess γ_Y is found from (5) and the half-width of the interval h_p is calculated from (6), (7), (12) or (13) as appropriate. The expanded uncertainty interval is $[y - h_p, y + h_p]$ where $y = F(x_1, \dots, x_m)$.

5. Extension for asymmetric distributions

So far, there has been no explicit consideration of input distributions that are asymmetric; we have worked only with variances and fourth cumulants, i.e. even cumulants. The basic procedure of the *Guide* also takes no explicit account of asymmetry, so our claim that the cumulants method is applicable wherever the basic procedure of the *Guide* is

applicable is not invalidated on this account. Nevertheless, we would like to be able to properly treat the asymmetric input distributions that arise in various branches of metrology. One example is the so-called ‘cosine error’ in length measurement. If L is the length of an object measured using a nominally parallel scale and if θ is the small angle between the axis of the object and the line of the scale then the measured length is $L \cos \theta$, which is necessarily less than L . A second example is found in the depressive effect of impurities on fixed points in the temperature scale, e.g. [28].

The cumulants method can be extended naturally to accommodate asymmetric distributions and distributions with non-zero mean. Equation (3) holds for all cumulants, so $\kappa_3(E_Y) = \sum_{i=1}^m c_i^3 \kappa_3(E_i)$ and $\kappa_1(E_Y) = \sum_{i=1}^m c_i \kappa_1(E_i)$. We simply obtain the mean, variance, coefficient of skewness and coefficient of excess of E_Y as $\kappa_1(E_Y)$, u_Y^2 , $\kappa_3(E_Y)/u_Y^3$ and γ_Y and then approximate the distribution of E_Y by the unique asymmetric Pearson distribution with these same four figures, i.e. the Pearson distribution with the same first four moments.

Suppose that we require the interval covering the central 100 $p\%$ of this distribution¹⁷. The limits of this interval are obtained from the $100(1-p)/2$ and $100(1+p)/2$ percentage points of the standardized form of this distribution by multiplying by the standard deviation u_Y and adding the mean, $\kappa_1(E_Y)$. Lower and upper percentage points of standardized Pearson distributions have been tabulated for certain probability levels and various coefficients of skewness and kurtosis [30, 31]. Such percentage points are calculable approximately for arbitrary probability levels [32, 33] or, if required, may be calculated to arbitrary precision from first principles¹⁸.

When the scale of the distribution has the distribution (14), the equation for the third cumulant analogous to (15) and (16) is

$$\kappa_3(E_i) = (1 + \epsilon^2) \kappa_3(E_i | \Theta = \delta).$$

A simple example analysis using an asymmetric distribution is given in appendix G.

6. Internal consistency and transferability

Consider, first, the method without the extension for asymmetric input distributions. Clause 0.4 of the *Guide* states that the quantity used to express uncertainty should be internally consistent, i.e.

it should be directly derivable from the components that contribute to it, as well as independent of how these components are grouped and of the decomposition of the components into subcomponents(.)

The output of the method can be seen to satisfy this criterion provided that the treatment of statistically correlated variables

¹⁷ This is not the shortest interval when the distribution is asymmetric [29]. Although the use of the shortest interval is advocated elsewhere in this paper, the ‘symmetric probability’ interval is preferred here because it enables the user to associate a known probability with each of the limits of the interval.

¹⁸ A future paper will provide equations similar to (6) and (7) to determine appropriate percentage points in the asymmetric case. These equations will involve both κ_3/u^3 and γ .

in their groups, which is appropriate, is not deemed to be a violation.

Similarly, clause 0.4 states that

it should be possible to use directly the uncertainty evaluated for one result as a component in evaluating the uncertainty of another measurement in which the first result is used.

The quantities u_Y^2 and γ_Y (or u_Y^2 and w_Y) are the only quantities that need to be recorded to satisfy this requirement of transferability.

Perhaps the measurand is to become an input to a further analysis that requires the uncertainty information in the form of a standard uncertainty u^* and a number of degrees of freedom ν^* . If these quantities are taken to be a scale factor and the degrees of freedom for a t -distribution possessed by the measurand (as implied by viewing the procedure of the *Guide* as a Bayesian procedure [15]), then these figures can be found when w_Y is positive because the Pearson approximation to the distribution of E_Y will itself be a scaled t -distribution. That is, if $w_Y > 0$ then

$$\nu^* = 4 + \frac{6u_Y^4}{w_Y}, \quad u^* = \left(\frac{\nu^*}{\nu^* - 2} \right)^{1/2} u_Y.$$

However, when $w_Y < 0$ the approximating distribution is a shifted and scaled beta-distribution, in which case the best we can do is set $u^* = u_Y$ and $\nu^* = \infty$. That is, the best scaled t -distribution for approximating the beta-distribution is a normal distribution.

Many practical measurements will result in a negative value of w_Y because most distributions used in Type B analyses have negative or zero excess. Only if there are significant contributions from distributions with positive excess, such as a t -distribution or a Laplace distribution might we find that $w_Y \geq 0$. The inability of the method of the *Guide* to assign a distribution with $w_Y < 0$ to the measurand is the basic reason for the widths in table 2 differing considerably from those of the Bayesian and cumulants methods when the uniform variable is dominant. A similar comment is applicable in the example of appendix A.

Now let us permit the presence of asymmetric input distributions. Evidently, the method is still internally consistent. Likewise, the results are transferable when the figures $\kappa_1(E_Y)$ and $\kappa_3(E_Y)/u_Y^3$ are recorded also. When wishing to calculate suitable figures of u^* and ν^* we can deal with the mean $\kappa_1(E_Y)$ by translation, but it is unclear how best to treat the skewness measure $\kappa_3(E_Y)/u_Y^3$. (This should not be seen as a limitation of the cumulants method but of the method that requires the uncertainty in the form (u^*, ν^*) and so takes no account of asymmetry.) Two possibilities that may be examined in further work are (i) to leave w_Y unaffected and treat the skewness as zero and (ii) to reduce w_Y by the amount $\kappa_3(E_Y)^2/u_Y^6$ and then treat the skewness as zero. The second course of action is suggested by inequality $\kappa_4/\kappa_2^2 > \kappa_3^2/\kappa_2^3 - 2$, which relates these measures of kurtosis and skewness for any distribution. (In the notation of appendix G, this is the known inequality $\beta_2 > \beta_1 + 1$ [34, exercise 3.19].)

7. Comments on the procedure of the *Guide* and the cumulants procedure

As implied in section 1, the procedure of the *Guide* is neither fully Bayesian nor frequentist. An intention to accommodate both understandings of probability is evident in its clause E.5.3, which states:

it makes no difference *in the calculations* if a standard uncertainty is viewed as a measure of the dispersion of the probability distribution of an input quantity or as a measure of the dispersion of the probability distribution of the error of that quantity.

(The italicization is that of the *Guide*.) The first view is Bayesian in nature, while the second accords with a frequentist understanding.

The chief example in the *Guide* of the mixing of ideas from these statistical frameworks is found in the frequentist evaluation of Type A uncertainty components, the Bayesian assignment of distributions to Type B uncertainty components and their combination by the Welch–Satterthwaite formula, which was derived under frequentist theory. If, instead, the prior distribution (8) were used in Type A evaluation and the Welch–Satterthwaite formula were not used, then the evaluation of uncertainty would become one of convolving distributions of uncertainty components in the linearized formulation. This convolution is the basis of the cumulants method.

The language of the *Guide* seems vague in places, perhaps because of the co-existence of the different philosophies. For example, clause 6.1.2 of the *Guide* states that

it is often necessary to give a measure of uncertainty that defines an interval about the measurement result that *may be expected* to encompass a large fraction of the distribution of values *that could reasonably be attributed* to the measurand.

(The italicization is ours.) The unique framework adopted here helps us to be more definite; we can say that the proposed interval is ‘an interval about the measurement result that *does* encompass a large *specified* fraction of a distribution of values that *can* reasonably be attributed to the measurand’. We have written ‘a distribution’ because different sets of reasonable assumptions will give equally valid output distributions.

Some of the *Guide*’s notation seems to obscure differences between the statistical frameworks. For example, in Type A evaluation the quantity $u(x_i)$ is equated with the experimental standard deviation of the sample mean $\bar{X}_i$ (see clause 4.2.3), but in Type B evaluation the same notation $u(x_i)$ is used to indicate the standard deviation of X_i (see clause 4.3.7); the note to clause 4.3.1 does acknowledge that the notation $u(X_i)$ is appropriate when a prior distribution has been assumed (as in Type B evaluation), but also indicates that the notation $u(x_i)$ is used for simplicity. Retaining the notation $u(X_i)$ would rightly indicate that $u(x_i)$ and $u(X_i)$ relate to different concepts and do not form a homogeneous set from which an overall standard uncertainty may be calculated easily or at all.

In this description of the cumulants procedure, we have maintained clarity by indicating quantities with pdfs in capitals and quantities that are fully known in lower case. Accordingly, the figure u_i associated with a measurement x_i is the standard

deviation of the (posterior) distribution of the unknown quantity X_i . The subscript notation is simpler than the function and argument notation $u(X_i)$.

The development of the *Guide* has been a major step forward in the standardization of the analysis of uncertainty. However, the author concurs wholeheartedly with the following paragraph of Gleser [14]¹⁹:

If the BIPM/ISO approach is followed, there will be international standardization of presentations of uncertainty, at least by national laboratories; such standardization is certainly highly desirable. The methods advocated for expressing uncertainty in the *Guide*, however, seem to form a new paradigm for statistical inference that is neither completely frequentist nor completely Bayesian, and consequently lack a firm theoretical basis. Thus, there is concern among statisticians that the methods advocated by the *Guide* could prove to be misleading or inaccurate. Widespread acceptance of the paradigm advocated in the *Guide* within the physical science community also has the potential to increase confusion about statistical concepts and thus impede communication between statisticians and nonstatisticians.

The cumulants procedure has been developed to address such concerns while remaining true to the direction and intent of the *Guide* as providing a practical, standardized and universal method for the evaluation of uncertainty.

8. Conclusion

The proposed procedure for the evaluation of measurement uncertainty is as simple as the basic procedure of the *Guide* and has most of the qualities of an ideal method, as outlined in clause 0.4 of the *Guide*. It allows the treatment of correlated input quantities with infinite and finite degrees of freedom and has greater flexibility than the procedure of the *Guide* in the representation of the distribution associated with the measurand. The method has been presented for the coverage probabilities of 0.95 and 0.99 but could easily be extended to include other specific coverage probabilities.

The procedure is based on established statistical theory, is reasonable and is transparent provided that (6), (7), (12) and (13) are understood to be close approximations to the relevant functions. The explicit choice of the statistical framework will assist the user to see how the procedure may be legitimately extended or modified in special circumstances. In particular, the method has a natural extension for use with input quantities with asymmetric distributions.

With or without this extension, the method involves approximating the distribution of the measurand by the unique distribution in the Pearson system with the same first four moments. The limits of the expanded uncertainty interval are easily found from the percentiles of the standard form of this Pearson distribution.

The Bayesian framework, which is utilized in the procedure and which is implied in other documents that build on the *Guide*, e.g. [16, 29], provides an intuitive

way of combining so-called random and systematic errors. However, adherence to this framework requires that the measurement estimate y , the standard uncertainty u_y , and the uncertainty interval $[y - h_p, y + h_p]$ be given Bayesian interpretations when they themselves are to be used in subsequent calculations; for example, in the calculation of a calibration curve, a degree of equivalence or a comparison reference value, e.g. [35]. So the acknowledgment that uncertainties are derived in a Bayesian manner is likely to have an impact on the validity of other statistical methods used in metrology, many of which are derived using frequentist theory.

In addition, many metrologists will regard the proper means of assessing the performance of a method of uncertainty analysis to involve repeated trials conducted at realistic fixed values of the unknown parameters, e.g. X_1 , Σ^2 in section 3 and X and V in section 4.4 [36]. In such an assessment, the Bayesian method with the prior pdfs (8) and (17) (which involves assuming that these pdfs represent the realistic distribution of the parameters) will, in general, fail to give the required proportion of uncertainty intervals containing the measurand, e.g. 95%²⁰. This issue brings us back to what may be the heart of the matter: the use of the 'non-informative' prior distributions (8) and (17) required to present the standard Bayesian method as an off-the-shelf means of uncertainty evaluation is making assumptions that many practising metrologists would not be inclined to accept.

These are issues that are well worth considering before the metrology community commits itself fully to the Bayesian framework. A rigorous theory of statistical inference that is somehow able to combine frequentist and Bayesian principles would be of great advantage.

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Appendix A. Application in example H1 of the *Guide*

The method was applied to the data of example H1 in Annex H of the *Guide*, which describes the evaluation of the uncertainty in the length of a gauge block by comparison with a standard. The half-width of a 99% interval for the length of the block was calculated to be 87.6 nm using the cumulants method. This is compared with the half-width of 93 nm obtained using the procedure of the *Guide* and stated in clause H.1.6.

The six sources of uncertainty contributing to the calculation are listed in table 4, which is based on table H.1 of the *Guide*. For simplicity, each partial derivative c_i has been incorporated into the variable E_i , so in effect $c_i = 1$. The variance u_i^2 and fourth cumulant w_i of each of these variables are obtained from the distributions that are now identified.

The first source of uncertainty relates to the calibration of the standard gauge. The calibration certificate gives an

²⁰ This is not immediately apparent with the usual Type A evaluation. However, it becomes so when there are correlated variables and (17) is used [36].

¹⁹ Gleser's paper indicates the subject to be the *Guide*'s 1993 printing.

Table 4. Summary of analysis for example H1 of the *Guide* for the cumulants method (using notation of this paper) and the method of the *Guide* (using notation of the *Guide*).

Source of uncertainty	<i>i</i>	Cumulants method		<i>Guide</i>	
		u_i^2/nm^2	w_i/nm^4	Term/nm	d.o.f.
Calibration of standard gauge	1	703.125	211879.2	25	18
Repeated observations	2	36.873	407.9	5.8	24
Random effects of comparator	3	25.214	3814.5	3.9	5
Systematic effects of comparator	4	45.370	0	6.7	8
Difference in expansion coefficients	5	8.361	−82.2	2.9	50
Difference in temperatures	6	298.481	−60602.9	16.6	2
Summary measure of scale		$u_Y = 33.4 \text{ nm}$		$u_c = 32 \text{ nm}$	
Summary measure of shape		$\gamma_Y = 0.124$		$\nu_{\text{eff}} = 16$	
Coverage factor for 99% probability		$k_{0.99}(\gamma_Y) = 2.62$		$t_{99,16} = 2.92$	
Expanded uncertainty		$h_{0.99} = 87.6 \text{ nm}$		$U_{99} = 93 \text{ nm}$	

expanded uncertainty of 75 nm obtained using a coverage factor of 3 and carrying a precision of 18 degrees of freedom. In accordance with the idea of assigning distributions to components, this is interpreted to mean that the corresponding error variable E_1 should be given the distribution of $T_{18} \times 75/3 \text{ nm}$.

The second error variable relates to ‘repeated observations’. Twenty-five observations of the difference in lengths of two standard gauges were used to estimate the standard deviation to be associated with any individual measurement, with the result being $s = 13 \text{ nm}$. The actual measurement involving the gauge of interest was the mean of five later measurements. So, from (11), the distribution assigned to the related error variable is that of $T_{24} \times 13/\sqrt{5} \text{ nm}$.

The third variable relates to random effects of the comparator. The calibration certificate of the comparator gives the uncertainty as $\pm 10 \text{ nm}$ at a level of confidence of 95% based on six repeated measurements. Analogously to E_1 , we set $E_3 \sim T_5 \times 10/2.571 \text{ nm}$.

The fourth variable relates to the systematic effects of the comparator. This has standard uncertainty ± 20 at the ‘three sigma level’, and this uncertainty is assumed to be reliable to 25% in clause H.1.6 of the *Guide*. In the absence of other information, we take the distribution of E_4 to be that of a normal variable with mean zero and standard deviation given by (14) with $\delta = 20/3$ and $\epsilon = 0.25$, i.e. $E_4 \sim N(0, \Theta^2)$ and $\Theta \sim U[0.75 \times 20/3, 1.25 \times 20/3] \text{ nm}$.

The fifth variable relates to the difference in the thermal expansion coefficients. This is a uniform variable with limits $\pm 10^{-6} \text{ }^\circ\text{C}^{-1}$ multiplied by the factor $0.05 \text{ m} \times 0.1 \text{ }^\circ\text{C}$ that is seen in equation (H.6a) of the *Guide*. The limits are deemed to be reliable to 10%. So $E_5 \sim U[-\Theta, \Theta]$ and $\Theta \sim U[0.9 \times 5, 1.1 \times 5] \text{ nm}$.

The sixth variable relates to the difference in temperatures. This is a uniform variable with limits $\pm 0.05 \text{ }^\circ\text{C}$ multiplied by the factor $0.05 \text{ m} \times 11.5 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ that is also seen in equation (H.6a) of the *Guide*. The limits are deemed to be reliable to only 50%. So $E_6 \sim U[-\Theta, \Theta]$ and $\Theta \sim U[0.5 \times 28.75, 1.5 \times 28.75] \text{ nm}$.

The values of u_i^2 and w_i for E_1 , E_2 and E_3 can be found from table 1. For E_4 , E_5 and E_6 we also require (15) and (16). The variances and fourth cumulants are added to obtain $u_Y^2 = 1117.424$ and $w_Y = 155416.5$. Then γ_Y is calculated, the multiplier $k_{99}(\gamma_Y)$ is obtained from (7) and the half-width $h_{99} = k_{99}(\gamma_Y)u_Y$ is calculated.

The measurement estimate y is unaffected by the method of uncertainty analysis, so we need only compare the half-widths of the intervals, which are 87.6 nm for the cumulants method and 93 nm for the method of the *Guide*. (The figure of 93 nm is larger than it need be because the *Guide* rounds down the number of degrees of freedom to 16. More precise results are $u_c = 31.7 \text{ nm}$, $\nu_{\text{eff}} = 16.75$, $t_{99}(\nu_{\text{eff}}) = 2.904$ and $U_{99} = 91.9 \text{ nm}$.) In this case the cumulants method has resulted in an interval that is approximately 5% narrower while having the same level of assurance, 99%. So the metrologist using the cumulants method will justifiably give the user a more precise statement about the measurand. This important concept is discussed further in appendix B.

As in table 2, the principal reason that the proposed method gives a narrower interval is the presence of the variables with negative kurtosis, especially E_6 . These counteract the variables with positive excess E_1 , E_2 and E_3 and make the distribution of E_Y much more normal, so making the tails of the distribution lighter. The standard deviations, $u_Y = 33.4$ and $u_c\sqrt{\nu_{\text{eff}}/(\nu_{\text{eff}} - 2)} = 33.7$, are similar. However, the shape parameters differ considerably; $\gamma_Y = 0.124$ corresponds to a t -distribution with approximately 52 degrees of freedom, whereas the t -distribution with $\nu_{\text{eff}} = 16.75$ has $\gamma \approx 0.47$.

Appendix B. Statistical performance criteria

Section 6 considered certain qualities that the *Guide* requires of a practical method of evaluating measurement uncertainty. The cumulants method possesses these qualities. This appendix discusses qualities of a more statistical nature, especially the idea alluded to in appendix A, that a procedure that produces narrower intervals while maintaining the quoted level of coverage is superior.

The author believes that any potential user of a large number of independent measurements each given a 95% uncertainty interval should be able to expect 95% of these intervals, in the limit, to contain the value of the relevant measurand²¹. The fidelity to, or exceeding of, this probability

²¹ This suggests the following definition of such an interval: a 95% expanded uncertainty interval for a measurand is one in a set of equally reliable independent intervals for (the same or) other measurands of which 95%, on average, contain the value of that measurand [37]. This definition makes no reference to the disputed concept of a probability or a probability distribution. It is neither uniquely frequentist, where the emphasis is on the intervals that might be calculated for the same single measurand, nor is it uniquely Bayesian. It is much more practical than either of the definitions implied in section 7.

enables the user to have the quoted level of certainty. This must be the first statistical criterion for an uncertainty procedure²². However, the procedure should exceed this probability by as little as possible, so that the resulting interval can be made as narrow as possible, thereby giving the most informative statement to the user²³. Thus, a procedure that produces narrower intervals while giving an interval that covers the measurand with at least the quoted probability is to be preferred.

Some metrologists might see a narrow interval as being undesirable because of doubt about the distributions assigned to the input quantities and because of an inclination to be conservative. However, there is no guarantee that a different procedure producing a wider interval has a higher probability of giving an interval containing the measurand. For example, we can construct an interval that has a probability of exactly 95% of containing X_1 in section 3.1 using the symmetry of the sampled distribution but not the knowledge of its normality. This interval is (almost always) wider than (10) because it does not make full use of the model assumptions. So the comparative widths of intervals produced by different procedures are also related to the efficiency with which the procedures make use of the available information.

Any doubt or conservatism about an input distribution should be encoded in the specification of that distribution, e.g. in choosing a large value of ϵ in section 4.3. Subsequently, a metrologist wishing to be conservative should calculate an uncertainty interval for a raised coverage probability, say 96%, but ascribe it the nominal coverage probability, 95%. Then, approximations and assumptions notwithstanding, the overall procedure can be given the statistical label ‘conservative’.

Appendix C. Comments on the Bayesian approach

The Bayesian analyst needs to be able to quantify his or her prior belief held about every unknown quantity. These same beliefs must be held by the recipient of a Bayesian interval (e.g. a client) who wishes to accept it fully. These requirements appear to complicate the matters of obtaining defensible results and communicating these results to a third party. In essence, this is why the author usually favours a method of analysis that keeps with frequentist (classical) principles, e.g. [12, 13, 39, 40].

However, the Bayesian idea of assigning distributions to unknown input quantities is appealing to many metrologists, and this framework leads to simple, intuitive, theory when combining these distributions. Also, if prior distributions can be agreed upon, a major objection of many holding the frequentist view is overcome, while it seems more difficult for those holding a Bayesian view to find meaning in a frequentist procedure. These ideas lend some support to the procedure in this paper, which may be viewed as a Bayesian procedure.

The concept of a fixed single value of any unknown quantity is not part of the Bayesian statistical framework.

²² Indeed, in frequentist statistics, the word ‘valid’ is used to describe a procedure with this property.

²³ Stevens [38] writes that ‘it is a statistician’s duty to be wrong the stated proportion of times, and failure to reach this proportion is equivalent to using an inefficient in place of an efficient method of estimation’. (The italics are those of Stevens.) The author agrees with this statement and regards it as fundamental.

In this framework the quantity is identified with, and is inseparable from, its distribution. So no particular point on the ‘x-axis’ of the distribution of the measurand has any intrinsic meaning. However, as the next paragraph shows, each of the mean, median or mode might be taken as the best point estimate of the value of the measurand.

Because the total knowledge about the measurand Y in a Bayesian analysis is described by its posterior pdf $f_Y(z)$, its best estimate y_b may be chosen by defining a loss-function, $l(z - y_b)$, and minimizing the expected loss $\int_{-\infty}^{\infty} f_Y(z) l(z - y_b) dz$, e.g. [41, chapter 1.6]. It turns out that if $l(x) \propto x^2$ then the best estimate is the mean of Y but if $l(x) \propto |x|$ then the best estimate is the median. If no loss-function is proposed then the best estimate might seem to be the value that is most likely, which is the mode, (but see [42]). If the distribution is symmetric and strictly unimodal, as with all symmetric Pearson distributions with $\gamma > -1.2$, then these three quantities are equal. So if our symmetric approximation to the distribution of Y has $\gamma_Y > -1.2$ then these quantities are each equal to the measurement estimate y .

Appendix D. Type A evaluation: derivation of (11)

Let us drop the subscript i from X_i . For consistency, we should use the same prior distributions for X and Σ^2 as those contributing to (8), which are the (improper) pdfs $f_X(x) = \text{constant}$ and $f_{\Sigma^2}(z) \propto 1/z$.

The prior distribution of X is uniform so, by application of (1), the posterior pdf of X on the condition that $\Sigma^2 = \sigma^2$ is proportional to the likelihood function evaluated at $\Sigma^2 = \sigma^2$.²⁴ Thus, if X is estimated by the mean of n observations, $\bar{x}$, then the posterior distribution of X on the condition that $\Sigma^2 = \sigma^2$ is normal with mean $\bar{x}$ and variance σ^2/n [22, p 72; 23, p 67], i.e. $X | (\Sigma^2 = \sigma^2) \sim N(\bar{x}, \sigma^2/n) \sim \bar{x} + \sigma/\sqrt{n} \times N(0, 1)$.

Also, if Σ^2 is estimated from a separate sample of k independent measurements with sample variance s^2 then the posterior distribution of Σ^2 is such that $(k-1)s^2/\Sigma^2$ has the pdf of a chi-square variable with $k-1$ degrees of freedom, i.e. $\Sigma^2 \sim (k-1)s^2 \chi_{k-1}^{-2}$ [22, p 71; 23, p 67–8].

The unconditional posterior distribution of $X - \bar{x}$ is therefore the distribution of a standard normal variable multiplied by $\sigma/\sqrt{n}$ and divided by the square root of an independent variable distributed as $\{\chi_{k-1}^2/(k-1)\}/s^2$. So the posterior distribution of X is given by (11).

Appendix E. Type A evaluation: input variables with $\nu \leq 4$

We demonstrate the performance of the modification in section 4.2 using the example of section 3 with $n = 3, 4, 5$, so $\nu = 2, 3, 4$. The half-widths of the intervals calculated by the cumulants method and the Bayesian method are given in table 5 for various values of s . The results are satisfactory. Differences can be observed when the scales of the variables are comparable, as in table 2. Some additional error seems present in table 5.

²⁴ The likelihood function is the joint pdf of the random data when it is evaluated at the observed data points and regarded as a function of the parameters. Here the parameters are X and Σ^2 , with $\Sigma^2 = \sigma^2$.

Table 5. Half-widths of the 95% and 99% uncertainty intervals for $Y = X_1 + X_2$.

n	r^*	95%				99%				ν_Y		
		Bayesian	Cumulants	A	B	Bayesian	Cumulants	A	B	Cumulants	A	B
3	0.01	0.48	0.48	0.48	0.48	0.50	0.50	0.50	0.50	NA	NA	NA
3	0.1	0.49	0.54	0.51	0.59	0.57	0.60	0.54	0.75	NA	NA	NA
3	0.3	0.60	0.73	0.68	0.76	0.99	1.13	0.95	1.34	5.2	13.4	3.3
3	1	1.33	1.35	1.24	1.35	2.91	2.85	2.28	2.98	2.3	3.0	2.1
3	3	3.76	3.76	3.69	3.76	8.61	8.57	8.23	8.63	2.0	2.1	2.0
4	0.1	0.48	0.50	0.50	0.50	0.54	0.53	0.53	0.53	NA	NA	NA
4	0.3	0.56	0.64	0.63	0.65	0.74	0.84	0.80	0.89	649	NA	15.7
4	1	1.05	1.13	1.15	1.12	1.77	1.88	1.80	1.91	4.0	5.1	3.6
4	3	2.80	2.85	2.88	2.84	5.09	5.15	5.14	5.15	3.1	3.2	3.1
5	0.1	0.48	0.49	0.49	0.49	0.53	0.51	0.51	0.52	NA	NA	NA
5	0.3	0.55	0.59	0.58	0.59	0.68	0.70	0.69	0.71	NA	NA	NA
5	1	0.95	1.00	1.00	1.00	1.45	1.54	1.52	1.55	5.5	5.9	5.2
5	3	2.46	2.48	2.49	2.48	4.03	4.06	4.01	4.07	4.2	4.5	4.1

Notes: Cumulants method: $u^2 = 27/\nu^2$ for $\nu < 3$, $\gamma = 150/\nu^2$ for $\nu < 5$. Variant A: $u^2 = 9/\nu$ for $\nu < 3$, $\gamma = 30/\nu$ for $\nu < 5$. Variant B: $u^2 = 81/\nu^3$ for $\nu < 3$, $\gamma = 750/\nu^3$ for $\nu < 5$. Values of ν_Y are noted where applicable. The ratio of scale factors is $r^* = (s/\sqrt{n}) \div (1/\sqrt{12})$.

Our approach of ascribing a finite value to an infinite moment may seem unreasonable. However, it is supported by the argument that infinite tails are not appropriate and also by the fact that inferring the distribution of E_Y from its low-order moments is an approximate step anyway, with the likely quality of the approximation being limited when a relevant moment is infinite. Admittedly, our choices of the relationships $u^2 = 27/\nu^2$ for $\nu < 3$ and $\gamma = 150/\nu^2$ for $\nu < 5$ are somewhat *ad hoc*, but the sensitivity of the resulting interval to these choices is much less than might be expected. If, instead, we use $u^2 = 9/\nu$ and $\gamma = 30/\nu$ then we obtain the results in column A of table 5, while if we use $u^2 = 81/\nu^3$ and $\gamma = 750/\nu^3$ then we obtain the results in column B. Both of these variant settings have the same properties of continuity as the actual choice. The half-width for the actual choice is equal to or lies between the results for the variants A and B. In general, the differences between the half-widths obtained under these three rules are not large.

Appendix F. Statistical correlation: derivation of (21)

This appendix justifies the use of (21) in place of (4) when two or more input variables are assessed in the same single set of repeated measurements. To assist here, we note that when the pdf of Σ^2 is $f_{\Sigma^2}(z) \propto 1/z$ the prior pdf of the alternative measure of dispersion $1/\Sigma^2$ is $f_{1/\Sigma^2}(1/v) \propto v$. So (8) is equivalent to the joint pdf

$$f_{X_1, 1/\Sigma^2}(x, 1/v) \propto v. \quad (22)$$

Let us assume for the moment that $X_1, \dots, X_p$ are p input quantities assessed in the same set of n independent measurements. We require the variance and fourth cumulant of $E \equiv \sum_{i=1}^p c_i E_i$ for combination with the figures for the other input quantities. Let $\bar{x} \equiv (\bar{x}_1, \dots, \bar{x}_p)'$ be the column vector of the sample means and $\bar{s} \equiv (s_{ij})$ be the $p \times p$ matrix of the sample covariances. (A bold italic letter denotes a column vector and a bold upright letter denotes a matrix. As before, the upper case indicates a random quantity).

If each sample vector $(x_{1,[k]}, \dots, x_{p,[k]})'$ is assumed to be an independent realization of a p -dimensional multivariate normal variable with mean vector $\mathbf{X} \equiv (X_1, \dots, X_p)'$ and covariance matrix $\mathbf{V} \equiv (V_{ij})$, and if our joint prior distribution for $\mathbf{X}$ and $\mathbf{V}^{-1}$ is (17), which is a generalization of (8) through the re-expression (22), then the posterior distribution of $\mathbf{X}$ is a (shifted and scaled) multivariate t -distribution with density function [43, equations (2.3) and (2.9)]

$$f_{\mathbf{X}}(\mathbf{x}) \propto \left\{ 1 + \frac{n(\mathbf{x} - \bar{\mathbf{x}})' \mathbf{s}^{-1} (\mathbf{x} - \bar{\mathbf{x}})}{n-1} \right\}^{-(n+p-1)/2}.$$

Consequently, for any non-null column vector $\mathbf{a}$ of length p , the scalar quantity $\mathbf{a}'(\mathbf{X} - \bar{\mathbf{x}})/\sqrt{\mathbf{a}'\mathbf{s}\mathbf{a}/n}$ has the (univariate) t -distribution with $n-1$ degrees of freedom [43, equation (3.7)]. For example, setting $\mathbf{a}$ to be the vector with i th element equal to 1 and all other elements equal to 0 gives the posterior distribution $X_i \sim \bar{x}_i + T_{n-1}\sqrt{s_{ii}/n}$, which agrees with (9)²⁵ because $s_{ii} = s_i^2$.

In the situation of interest, $\mathbf{a}$ is the vector $\mathbf{c} \equiv (c_1, \dots, c_p)'$ and $\mathbf{X} - \bar{\mathbf{x}}$ is the vector $(E_1, \dots, E_p)'$, so $\mathbf{a}'(\mathbf{X} - \bar{\mathbf{x}}) = E$. Thus

$$E \sim (\mathbf{c}'\mathbf{s}\mathbf{c}/n)^{1/2} T_{n-1} = \left(\sum_{i=1}^p \sum_{j=1}^p c_i c_j s_{ij} / n \right)^{1/2} T_{n-1},$$

from which we find that the variance and fourth cumulant of E are $u^2 = \sum_{i=1}^p \sum_{j=1}^p c_i c_j u_{ij}$ and $w = 6u^4/(n-5)$. (An equivalent expression for u^2 is $\sum_{i=1}^p c_i^2 u_i^2 + 2 \sum_{i=1}^{p-1} \sum_{j=i+1}^p c_i c_j u_{ij}$.)

So the group of p input quantities can be thought of as contributing a single independent error variable E with variance u^2 and fourth moment w . If we return to our general case of m input variables and a set of indices A we obtain (20), which leads to (21).

²⁵ Although (8) and (22) both describe the univariate Jeffreys prior, (17) is not the multivariate Jeffreys prior. The multivariate Jeffreys prior is $f_{\mathbf{X}, \mathbf{V}}(\mathbf{x}, \mathbf{v}) \propto |\mathbf{v}|^{-(p+1)/2}$. Using this would lead to a multivariate t -distribution with $n-p$ degrees of freedom [23, p 81]. The resulting univariate posterior distribution of X_i would not be (9), so the method would be inconsistent.

Table 6. 95% and 99% uncertainty intervals for $Y = X_1 + X_2$ with X_2 distributed exponentially with parameter $b = 1/\sqrt{12}$.

n	r	95%		99%		E_Y	
		Bayesian	Cumulants	Bayesian	Cumulants	$\sqrt{\beta_1}$	β_2
6	0.1	0.30, 0.78	0.29, 0.78	0.32, 1.24	0.30, 1.24	1.97	8.88
6	0.3	0.36, 0.79	0.34, 0.78	0.45, 1.25	0.37, 1.24	1.76	8.09
6	1	0.69, 0.94	0.72, 0.90	1.01, 1.43	1.00, 1.40	0.71	6.00
6	3	1.78, 1.85	1.81, 1.83	2.76, 2.82	2.80, 2.86	0.06	7.92
6	10	5.77, 5.78	5.78, 5.78	9.04, 9.04	9.05, 9.06	0.00	8.88
12	0.1	0.30, 0.78	0.29, 0.78	0.32, 1.24	0.30, 1.24	1.97	8.88
12	0.3	0.37, 0.79	0.34, 0.78	0.44, 1.25	0.36, 1.24	1.76	8.06
12	1	0.70, 0.93	0.70, 0.91	0.94, 1.40	0.91, 1.37	0.71	4.71
12	3	1.79, 1.84	1.79, 1.84	2.50, 2.59	2.50, 2.59	0.06	3.75
12	10	5.77, 5.78	5.77, 5.78	8.14, 8.14	8.14, 8.14	0.00	3.84

Notes: A paired entry h_1, h_2 represents the interval $[\bar{x} + b - h_1, \bar{x} + b + h_2]$. The figure $r = (s/\sqrt{n})\sqrt{(n-1)/(n-3)} \div b$ is the ratio of the standard deviation of X_1 to that of X_2 .

Appendix G. Example with an asymmetric input distribution

Suppose that the example in section 3 is modified so that X_2 is now known only to be positive. Let b denote its estimate again so $E_2 = X_2 - b$. Following the recommendation of the supplement to the Guide [16, clause 4.1] we assign X_2 the exponential pdf $f(x) = \exp(-x/b)/b$ for $x > 0$. Then $\kappa_1(E_2) = 0$, $\kappa_2(E_2) = b^2$, $\kappa_3(E_2) = 2b^3$ and $\kappa_4(E_2) = 6b^4$ [44, p 498]. Again $E_Y = E_1 + E_2$ so

$$\begin{aligned}\kappa_1(E_Y) &= 0, \\ u_Y^2 &= \frac{s^2(n-1)}{n(n-3)} + b^2, \\ \kappa_3(E_Y) &= 2b^3, \\ \gamma_Y &= \left\{ \frac{6s^4(n-1)^2}{n^4(n-3)^2(n-5)} + 6b^4 \right\} \div u_Y^4.\end{aligned}$$

The notation $\sqrt{\beta_1}$ is standard for the relevant dimensionless coefficient of skewness, which is a signed quantity, and the notation β_2 is standard for the relevant dimensionless coefficient of kurtosis, which is the coefficient of excess plus 3. The values of $\sqrt{\beta_1}(E_Y) = \kappa_3(E_Y)/u_Y^3$ and $\beta_2(E_Y) = \gamma_Y + 3$ were calculated for various pairs of n and s/b , and the 95% and 99% uncertainty intervals for Y were found using bivariate ‘four-point’ interpolation [45, equation (25.2.66)] in Table 32 of [31], which has $\sqrt{\beta_1}$ and β_2 as the marginal entries. Table 6 gives figures defining these intervals and the intervals calculated by the full Bayesian method. We chose $b = 1/\sqrt{12}$ so that the figures h_1 and h_2 in any row of table 6 are both directly comparable with the half-width in the corresponding row of table 2 (where the standard deviation of X_2 was $1/\sqrt{12}$). The values of $\sqrt{\beta_1}(E_Y)$ and $\beta_2(E_Y)$ are also tabulated²⁶.

Agreement between the cumulants method and the Bayesian method is good, with the exception of the 99% interval in the second row. As in table 2, there are differences where the variables have comparable standard deviations, but these differences would diminish as more variables became involved. The results would be exact in the theoretical limit of $r \rightarrow 0$ because the exponential distribution is a member of the Pearson family.

²⁶ As implied, $\sqrt{\beta_1}$ can be a negative quantity. It happens that $\sqrt{\beta_1}$ is always positive in this example.

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